

Single Mean Set 22 · one sample, σ unknown

Look for: "mean/average", "one group/sample", "Is the mean greater/less than X?", "estimate the mean μ"
"σ=1.4 is known for this process..." → Case I | "Survey of 50 households, s=\$85" → n≥30, σ unknown → Case II | "Sample of 10 patients, s=1.4" → n<30 → Case III

Case I — σ known (any n)

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \quad \text{Cl: } \bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Case II — σ unknown, n ≥ 30

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim N(0, 1) \quad \text{Cl: } \bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Case III — σ unknown, n < 30, normal population

$$T = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1} \quad \text{df} = n - 1 \quad \text{Cl: } \bar{x} \pm$$

$$t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}$$

Two Independent Means Sets 24–25

Look for: "compare means of two separate groups", "do A and B differ in average?", "two independent samples" — **not** the same people measured twice

σ₁, σ₂ given → Case I | n₁=50, n₂=45, σ unknown → Case II | small n, s ratio ≤1.4 → Case III | s ratio >1.4 → Case IV

Case I — σ₁, σ₂ known (any n)

$$Z = \frac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \sim N(0, 1) \quad \text{Cl: } (\bar{x} - \bar{y}) \pm z_{\alpha/2} \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$$

Case II — σ unknown, n₁≥30 & n₂≥30

$$Z = \frac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} \sim N(0, 1) \quad \text{Cl: } (\bar{x} - \bar{y}) \pm z_{\alpha/2} \sqrt{s_1^2/n_1 + s_2^2/n_2}$$

Case III — small n, ratio ≤ 1.4 (equal var, Pooled T)

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$T = \frac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)_0}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{n_1 + n_2 - 2}$$

$$\text{Cl: } (\bar{x} - \bar{y}) \pm t_{\alpha/2, \nu} \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

Critical values (c.v.):

Two-tail: 90%→1.645 · 95%→1.96 · 99%→2.576

One-tail: 90%→1.28 · 95%→1.645 · 99%→2.33

Single Proportion Set 23

$$\hat{p} = x/n \cdot e.s.e.(CI) = \sqrt{\hat{p}(1 - \hat{p})/n}$$

Look for: "proportion/% who...", "more than half...", yes/no outcome, one group, "Is the rate greater than p?"
"200 voters: 54% support bill. Evidence >50% support?" → p₀=0.5, one-tail test | "80 items, 12 defective. Estimate true defect rate with 90% CI." → CI (use $\hat{p}$, not p₀)

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} \sim N(0, 1) \quad \text{test} \rightarrow \text{use } p_0$$

$$\text{Cl: } \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

$$n = \frac{z_{\alpha/2}^2 \cdot \hat{p}(1 - \hat{p})}{E^2} \quad [\hat{p} = 0.5 \text{ if unknown}]$$

Paired Data Set 26 · same subjects, two treatments

Look for: "before & after", "same subjects measured twice", "matched pairs", "each participant received both treatments"
"10 runners timed before and after training. Did training reduce mean time?" → D = before–after | "BP measured morning & evening for 8 patients. Is there a difference?" → same person = paired, not 2-sample

1. Compute: $D_i = X_i - Y_i$ (e.g. after – before)

2. Find $\bar{d}$ and s_D ← calculator 1-var stats on the D_i values ($\bar{x} = \bar{d}$, $s_x = s_D$)

Usually $(\mu_D)_0 = 0$ ("no difference"). Non-zero only if claim says "by k" — e.g. "reduces by 5" → $(\mu_D)_0 = 5$.

Case I — n ≥ 30 (large number of pairs)

$$3. Z = \frac{\bar{d} - (\mu_D)_0}{s_D/\sqrt{n}} \sim N(0, 1) \quad \text{Cl: } \bar{d} \pm z_{\alpha/2} \cdot \frac{s_D}{\sqrt{n}}$$

Case II — n < 30, differences are normal

$$3. T = \frac{\bar{d} - (\mu_D)_0}{s_D/\sqrt{n}} \sim t_{n-1} \quad \text{Cl: } \bar{d} \pm t_{\alpha/2, n-1} \cdot \frac{s_D}{\sqrt{n}}$$

Regression & Correlation Sets 28–30 · $\hat{Y} = b_0 + b_1x$ · compute s_{xy} , s_x^2 , s_y^2 first

Look for: "predict Y from X", "linear relationship between X and Y", "for each unit increase in X, Y changes by...", "correlation"
"Hours studied (x) vs exam score (y). Find regression line; test if x is a significant predictor." · "Interpret b₁: each extra hour → score ↑ by b₁ pts" · "Predict score when x=6."

Variables: x = predictor (independent/explanatory) · y = response (dependent/outcome) · $\hat{y}$ = predicted y · b₀ = y-intercept (y when x=0) · b₁ = slope (change in y per +1 x) · β₁ = true slope · **e = y – $\hat{y}$** = residual (actual – predicted)

s_{xy} = sample covariance of x & y · **s_x², s_y²** = sample variances · **r** = correlation (–1 ≤ r ≤ 1, strength & direction) · **p** = true correlation · **R²** = % of variation in y explained by x · **SSE** = sum of squared residuals · **s_e** = std error of regression (avg distance from line)

$$s_{xy} = \frac{\sum x_i y_i - \left(\frac{(\sum x_i)(\sum y_i)}{n} \right)}{n - 1} \quad \text{compute } \sum x_i y_i \text{ by hand}$$

$$s_x^2 = \frac{\sum x_i^2 - \left(\frac{(\sum x_i)^2}{n} \right)}{n - 1} \quad \text{same for } s_y^2$$

$$b_1 = s_{xy}/s_x^2 \cdot b_0 = \bar{y} - b_1 \bar{x} \quad \text{residual: } e = y - \hat{y}$$

$$\text{SSE} = (n - 1) \left(s_y^2 - \frac{s_{xy}^2}{s_x^2} \right) \quad s_e = \sqrt{\frac{\text{SSE}}{n - 2}}$$

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p-value from t-table (same tail rules, bracket t_{obs} by df):

e.g. t_{obs} = 2.49, df = 11:

t_{0.025, 11} = 2.201 < 2.49 < t_{0.01, 11} = 2.718

→ 0.01 < p < 0.025 (one-tail)

Two Proportions Set 27

pool $\hat{p}_p$ for test only; unpooled $\hat{p}$ for CI

Look for: "compare proportions of two groups", two yes/no outcomes, "difference in rates/percentages between A and B"

"80 men, 24 smoke; 100 women, 18 smoke. Is smoking rate higher for men?" → test (pool $\hat{p}_p$) | "Find 95% CI for difference in support rates." → CI (no pooling)

$\hat{p}_p = (x_1 + x_2)/(n_1 + n_2)$ ← H-test only

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_p(1 - \hat{p}_p) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim N(0, 1)$$

$$\text{Cl: } (\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

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$$r = s_{xy}/(s_x \cdot s_y) \quad R^2 = r^2$$

$$b_1 = \frac{s_{xy}}{\sqrt{(n - 1)s_x^2}} \quad \text{e.s.e. of } b_1$$

T (slope): T = b₁/s_{b1} · t_{n–2} · H₀: β₁ = 0 vs H₁: β₁ ≠ 0

T (corr): T = r√(n–2)/(1–r²) · t_{n–2} · H₀: ρ = 0

Traps:

CI prop → $\hat{p}$; test → p₀

2-prop test → pool $\hat{p}_p$; CI → no pool

(μ₁ – μ₂)₀ = 0 unless claim says "by k"

Before/after → paired, not 2-sample

p-value from z-table (table gives P(Z < z)):

H₁: θ > val → p = 1 – table(z_{obs}) upper tail

H₁: θ < val → p = table(z_{obs}) lower tail

H₁: θ ≠ val → p = 2(1 – table(|z_{obs}|)) two-tail